Spring 2014

Marmara University
Electrical and Electronics Engineering
STAT253-Probability and Statistics
Final Exam

1. Let the discrete random variables X and Y have the joint probability mass function (PMF)

$$p_{X,Y}[i,j] = \frac{i+j}{32} \qquad i = 1,2, \quad j = 1,2,3,4$$

   (a) Find the expected values and variances of X and Y. [10 pts.]
   (b) Calculate the correlation coefficient. [10 pts.]
   (c) Are X and Y independent? [5 pts.]

2. Let the continuous random variables X and Y have the joint probability density function (PDF)

$$p_{X,Y}(x,y) = \begin{cases} 2, & 0 \le y \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

   (a) Calculate the marginal PDFs $p_X(x)$ and $p_Y(y)$. [10 pts.]
   (b) Find the expected values and variances of X and Y. [10 pts.]
   (c) Calculate cov(X,Y). [10 pts.]

3. Let the discrete random variables X and Y have the following joint PMF

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$$p_{X,Y}[i,j] = \begin{cases} \frac{1}{16}, & 1 \le i = j \le 4 \\ \frac{2}{16}, & 1 \le i < j \le 4 \end{cases}$$

   (a) Compute $E[Y|X=1]$ and VAR$(Y|X=1)$. [20 pts.]
   (b) Calculate $P(Y \le 2|X=2)$ and $P(X=2|Y=3)$. [10 pts.]

4. A random variable X is known to have $E[X] = 17$ and $E[X^2] = 298$. By using Chebyshev inequality

   (a) Obtain a lower bound for $P(10 < X < 24)$. [10 pts.]
   (b) Give an upper bound for $P(|X-17| \ge 16)$. [5 pts.]

Directions:

- One A4-sized formula sheet (without solutions) can be used.
- The exam is 75 minutes.

GOOD LUCK!

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FINAL EXAM - Solutions

1) $P_{X,Y}[i,j] = \frac{i+j}{32}$ $\quad i=1,2 \quad j=1,2,3,4$

i \ j	1	2	3	4
1	2/32	3/32	4/32	5/32
2	3/32	4/32	5/32	6/32

(a)

i	1	2
$P_X[X=i]$	$\frac{14}{32}$	$\frac{18}{32}$

j	1	2	3	4
$P_Y[Y=j]$	$\frac{5}{32}$	$\frac{7}{32}$	$\frac{9}{32}$	$\frac{11}{32}$

$$E[X] = \sum_{i=1}^{2} i\,P_X[i] = 1\cdot\frac{14}{32} + 2\cdot\frac{18}{32} = \frac{50}{32} \quad (2.5)$$

$$E[Y] = \sum_{j=1}^{4} j\,P_Y[j] = 1\cdot\frac{5}{32} + 2\cdot\frac{7}{32} + 3\cdot\frac{9}{32} + 4\cdot\frac{11}{32} = \frac{90}{32} \quad (2.5)$$

$$E[X^2] = \sum_{i=1}^{2} i^2 P_X[i] = 1^2\cdot\frac{14}{32} + 2^2\cdot\frac{18}{32} = \frac{86}{32}$$

$$E[Y^2] = \sum_{j=1}^{4} j^2\cdot P_Y[j] = 1^2\cdot\frac{5}{32} + 2^2\cdot\frac{7}{32} + 3^2\cdot\frac{9}{32} + 4^2\cdot\frac{11}{32} = \frac{370}{32}$$

$$\mathrm{var}(X) = E[X^2] - (E[X])^2 = \frac{86}{32} - \left(\frac{50}{32}\right)^2 \quad (2.5)$$

$$\mathrm{var}(Y) = E[Y^2] - (E[Y])^2 = \frac{370}{32} - \left(\frac{90}{32}\right)^2 \quad (2.5)$$

(b) $$\rho_{X,Y} = \frac{E[XY] - E[X]\,E[Y]}{\sqrt{\mathrm{var}(X)\,\mathrm{var}(Y)}}$$

$$E[XY] = \sum_{i=1}^{2}\sum_{j=1}^{4} ij\,P_{X,Y}[i,j] = \left(1.1\cdot\frac{2}{32} + 1.2\,\frac{3}{32} + 1.3\,\frac{4}{32} + 1.4\,\frac{5}{32}\right) + \left(2.1\cdot\frac{3}{32} + 2.2\cdot\frac{4}{32} + 2.3\cdot\frac{5}{32} + 2.4\cdot\frac{6}{32}\right)$$

$$= \frac{132}{32} \quad (5)$$

$$\rho_{X,Y} = \frac{\frac{132}{32} - \frac{50}{32}\cdot\frac{90}{32}}{\sqrt{\left[\frac{86}{32} - \left(\frac{50}{32}\right)^2\right]}\sqrt{\frac{370}{32} - \left(\frac{90}{32}\right)^2}} \quad (5)$$

(c) $P_{X,Y}[1,1] = \frac{2}{32}$ but $P_X[1]\cdot P_Y[1] = \frac{14}{32}\cdot\frac{5}{32}$ (5)

Hence, $P_{X,Y}[1,1] \neq P_X[1]\cdot P_Y[1] \Rightarrow X$ and Y are not independent.

2) (a) $P_X(x) = \int_{-\infty}^{\infty} P_{X,Y}(x,y)\,dy = \int_0^x 2\,dy = 2y\Big|_0^x = 2x$, $0 \le x \le 1$ (5)

$P_Y(y) = \int_{-\infty}^{\infty} P_{X,Y}(x,y)\,dx = \int_y^1 2\,dx = 2x\Big|_y^1 = 2-2y$, $0 \le y \le 1$ (5)

(b) $E[X] = \int_{-\infty}^{\infty} x\,P_X(x)\,dx = \int_0^1 x \cdot 2x\,dx = \frac{2x^3}{3}\Big|_0^1 = \frac{2}{3}$ (2.5)

$E[Y] = \int_{-\infty}^{\infty} y\,P_Y(y)\,dy = \int_0^1 y(2-2y)\,dy = y^2 - \frac{2}{3}y^3\Big|_0^1 = \frac{1}{3}$. (2.5)

$E[X^2] = \int_{-\infty}^{\infty} x^2 \cdot P_X(x)\,dx = \int_0^1 x^2 \cdot 2x\,dx = \frac{2x^4}{4}\Big|_0^1 = \frac{2}{4}$

$E[Y^2] = \int_{-\infty}^{\infty} y^2 \cdot P_Y(y)\,dy = \int_0^1 y^2(2-2y)\,dy = \frac{2y^3}{3} - \frac{2y^4}{4}\Big|_0^1 = \frac{2}{3} - \frac{2}{4} = \frac{2}{12}$

$Var(X) = E[X^2] - (E[X])^2 = \frac{2}{4} - \left(\frac{2}{3}\right)^2 = \frac{2}{4} - \frac{4}{9} = \frac{2}{36}$ (2.5)

$Var(Y) = E[Y^2] - (E[Y])^2 = \frac{2}{12} - \left(\frac{1}{3}\right)^2 = \frac{2}{12} - \frac{1}{9} = \frac{2}{36}$. (2.5)

(c) $Cov(X,Y) = E[XY] - E[X]E[Y]$

$E[XY] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\,P_{X,Y}(x,y)\,dx\,dy = \int_0^1\int_0^x xy\,2\,dy\,dx = 2\int_0^1 x\left[\int_0^x y\,dy\right]dx$ (5)

$= 2\int_0^1 \left(x \cdot \frac{y^2}{2}\Big|_0^x\right)dx = 2\int_0^1 x \cdot \frac{x^2}{2}\,dx = \frac{x^4}{4}\Big|_0^1 = \frac{1}{4}$.

$Cov(X,Y) = \frac{1}{4} - \frac{2}{3}\cdot\frac{1}{3} = \frac{1}{4} - \frac{2}{9} = \frac{1}{36}$. (5)

3)

i \ J	1	2	3	4
1	1/16	2/16	2/16	2/16
2	0	1/16	2/16	2/16
3	0	0	1/16	2/16
4	0	0	0	1/16

$\Rightarrow$

i	1	2	3	4
$P_X(i)$	7/16	5/16	3/16	1/16

J	1	2	3	4
$P_Y(J)$	1/16	3/16	5/16	7/16

(a) $E[Y|X=1] = \sum_{J=1}^{4} J \cdot P_{Y|X}(J|X=1)$

$P_{Y|X}(J=1|X=1) = \frac{1/16}{7/16} = \frac{1}{7}$, $P_{Y|X}(J=2|X=1) = \frac{2/16}{7/16} = \frac{2}{7}$ (5)

$P_{Y|X}(J=3|X=1) = \frac{2/16}{7/16} = \frac{2}{7}$, $P_{Y|X}(J=4|X=1) = \frac{2/16}{7/16} = \frac{2}{7}$

$E[Y|X=1] = 1\cdot\frac{1}{7} + 2\cdot\frac{2}{7} + 3\cdot\frac{2}{7} + 4\cdot\frac{2}{7} = \frac{19}{7}$ (5)

$\text{var}(Y|X=1) = E[Y^2|X=1] - (E[Y|X=1])^2$

$E[Y^2|X=1] = \sum_{j=1}^{4} j^2 \cdot P_{X|Y}(j|X=1) = 1^2 \cdot \frac{1}{7} + 2^2 \cdot \frac{2}{7} + 3^2 \cdot \frac{2}{7} + 4^2 \cdot \frac{2}{7} = \frac{59}{7}$

$\text{var}(Y|X=1) = \frac{59}{7} - \left(\frac{19}{7}\right)^2 = 1.06$ (10)

(b) $P(Y \le 2|X=2) = P(Y=1|X=2) + P(Y=2|X=2)$ (5)

$= \frac{0}{5/16} + \frac{1/16}{5/16} = \frac{1}{5}.$

$P(X=2|Y=3) = \frac{P_{X,Y}(X=2, Y=3)}{P_Y(Y=3)} = \frac{2/16}{5/16} = \frac{2}{5}.$ (5)

4) $\text{var}(X) = E[X^2] - (E[X])^2 = 298 - 17^2 = 298 - 289 = 9.$

(a) $P[|X-m| > \delta] \le \frac{\text{var}(X)}{\delta^2} \Rightarrow P[|X-17| > \delta] \le \frac{9}{\delta^2}$

(5)

$P[10 < X < 24] = P[|X-17| < 7]$ (5)

$P[|X-17| < 7] + P[|X-17| \ge 7] = 1 \Rightarrow P[|X-17| < 7] \ge 1 - \frac{9}{49} = \frac{40}{49}$

$\Downarrow$

$\le \frac{9}{49}$ (from chebyshev inequality!)

$\therefore$ Hence, $P[10 < X < 24] \ge \frac{40}{49}.$

(b) $P[|X-17| \ge 16] \le \frac{\text{var}(X)}{16^2} = \frac{9}{256}.$ (5)